

BAYESIAN SKIP-AUTOENCODERS FOR UNSUPERVISED HYPERINTENSE ANOMALY DETECTION IN HIGH RESOLUTION BRAIN MRI

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ABSTRACT

Autoencoder-based approaches for Unsupervised Anomaly Detection (UAD) in brain MRI have recently gained a lot of attention and have shown promising performance. However, brain MR images are particularly complex and require large model capacity for learning a proper reconstruction, which existing methods encounter by restricting themselves to downsampled data or anatomical subregions. In this work, we show that models with limited capacity can be trained and used for UAD in full brain MR images at their native resolution by introducing skip-connections, a concept which has already proven beneficial for biomedical image segmentation and image-to-image translation, and a dropout-based mechanism to prevent the model from learning an identity mapping. In an ablative study on two different pathologies we show considerable improvements over State-of-the-Art Autoencoder-based UAD models. The stochastic nature of the model also allows to investigate epistemic uncertainty in our so-called Skip-Autoencoder, which is briefly portrayed.

Index Terms— Anomaly Detection, Skip-Autoencoders, Uncertainty

1. INTRODUCTION

Deep unsupervised representation learning and deep generative modeling have recently turned out to be promising tools for Unsupervised Anomaly Detection (UAD) in 2D and 3D data. Autoencoders (AEs) and Generative models such as GANs or VAEs, respectively, have been successfully used to model normal anatomy. Since such models have only seen normality, anomalies will not properly be reconstructed and thus can be detected directly in image-space from high reconstruction errors. In general, UAD in brain MRI is of great interest for the Medical Image Analysis community, as it relieves from the need for pixel-level annotations—an important, but expensive and often scarce prerequisite for supervised training of deep neural networks—and allows to detect arbitrary pathologies without a-priori knowing about their appearance.

In the medical field, numerous works have been devoted to UAD in different modalities. In initial work, Schlegl et al. [1] proposed GANs for detecting anomalies in retinal OCT data, however at a level of small image patches. Baur et al. [2] introduced a feed-forward solution for UAD in brain MRI at high resolution with a combination of spatial AEs and GANs. While this method showed the ability to detect small MS lesions in high resolution data, it was limited to a narrow field of view centered around the axial midline. Another interesting concept based on bayesian convolutional AEs was proposed by Pawlowski et al. [3] to detect anomalies in head CT images, but their approach was again limited to few mid-axial slices. Zimmerer et al. [4] recently formulated UAD as an inpainting problem with their context variational AE, but validated their method at a resolution of only 64×64 px. Chen et al. [5] proposed constrained adversarial AEs for detecting large lesions in brain MR images, and there have also been first results on UAD in volumetric head CT data using 3D AEs [6]. However, both approaches were also limited to low resolution, which comes at the cost of information loss. This might be critical in the presence of small anomalous structures such as MS lesions. We hypothesize that the restriction to low-resolution and anatomical subregions are coping mechanisms to trade-off the limited model capacity, in particularly the dimensionality of the latent space, and the high complexity of brain MR data. For example, in brain MRI it is especially important to properly reconstruct fine details such as gyri und sulci, as they would otherwise falsely be recognized as anomalies.

Contribution: Motivated by their success in supervised biomedical image segmentation [7, 8] and image-to-image translation [9], we propose skip-connections in AEs to enable high fidelity reconstructions of complex, high resolution medical data. To prevent the model from learning an identity mapping and simply bypassing lower levels of the AE, we introduce a dropout-based identity mitigation mechanism. In an ablative study involving two different pathologies (Multiple Sclerosis and Glioblastoma), we show considerable improvements over standard AE architectures in the task of detecting

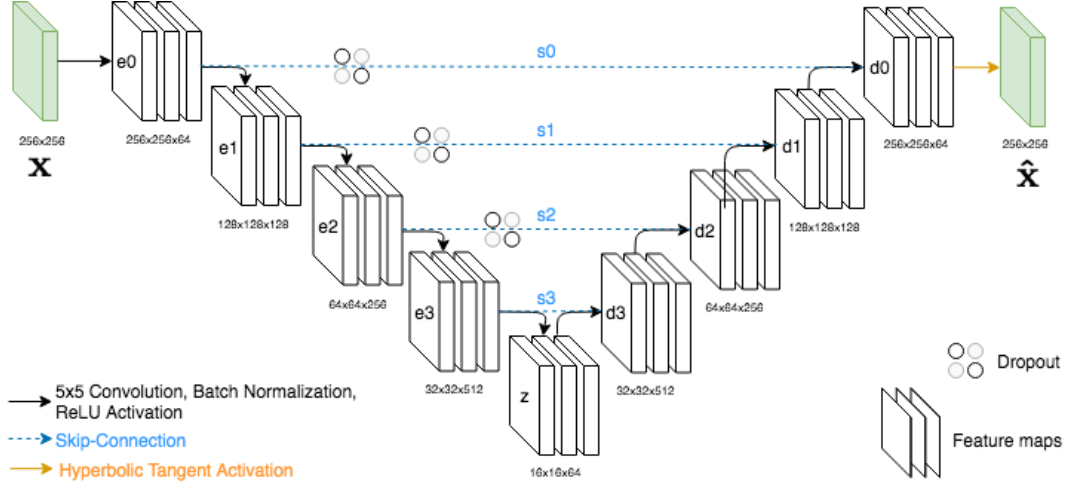


Fig. 1. The proposed Skip-Autoencoder architecture at a glance.

hyperintense anomalies in full resolution brain MR data purely from reconstruction errors. Further, since the model effectively is a Bayesian Neural Network, we briefly depict its epistemic uncertainty on normal and abnormal tissue.

2. METHODOLOGY

2.1. Modeling healthy anatomy with Skip-Autoencoders

Similar to previous work, we aim to model normal brain anatomy with an Autoencoder (AE) and detect anomalies from reconstruction residuals directly in 2D image space. To achieve this for complex medical data such as MRI without imposing constraints on resolution or field-of-view, we propose the so-called Skip-Autoencoder (skipAE): We introduce skip-connections in AEs (see Fig. 1) and train a model $f_{\theta}(\mathbf{x})$ to compress and reconstruct only anomaly-free data $\mathbf{x} \in \mathcal{X}_N$. We optimize the model's parameters θ with the loss

$$\mathcal{L}(\mathbf{x}, \hat{\mathbf{x}}) = \ell_1(\mathbf{x}, \hat{\mathbf{x}}) + \text{gdl}(\mathbf{x}, \hat{\mathbf{x}}) \quad (1)$$

where ℓ_1 is the ℓ_1 -distance between the input $\mathbf{x}$ and its reconstruction $\hat{\mathbf{x}}$, and $\text{gdl}(\cdot, \cdot)$ is the gradient difference loss which forces the model to reconstruct image gradients well [10].

Skip-connections have proven very useful in biomedical image segmentation [7] and image restoration [11] as they enhance gradient flow and provide a view on image features at multiple scales, leveraging both local and global features in the segmentation process. Their importance has been thoroughly studied in [8]. Recently, they have also proven beneficial in GAN-based image-to-image translation [9]. There, the authors employed a stochastic U-Net like architecture as the generator and identified skip-connections as a key component to obtain high fidelity results.

2.2. Mitigating Identity

The introduction of skip-connections in AEs theoretically allows the optimization to bypass the compressed representation of data and instead to learn an identity mapping. This would give rise to perfect reconstructions for any kind of data, which is not desirable for reconstruction-based UAD. With convolutional kernels of size $k \times k$, $k > 0$ and a bias term b , an identity mapping is achieved if for every value x_i of an input feature map $\mathbf{x}$ the following equation holds

$$w_{\lfloor \frac{k}{2} \rfloor, \lfloor \frac{k}{2} \rfloor} x_i + b = x_i \quad (2)$$

with $w_{\lfloor \frac{k}{2} \rfloor, \lfloor \frac{k}{2} \rfloor}$ being the central weight of the kernel and b being the bias. The trivial solution to this is $w_{\lfloor \frac{k}{2} \rfloor, \lfloor \frac{k}{2} \rfloor} = 1$ and $b = 0$, while all the other weights of the kernel are set to zero. To mitigate this, we i) initialize the weights randomly and ii) apply dropout to every skip-connection with a dropout-rate d , such that the central weight of a kernel is likely to be dropped every once in a while and the contribution of the other weights is enforced.

2.3. Bayesian Skip-Autoencoders

By applying dropout, we implicitly place a Bernoulli distribution over the network parameters θ , i.e. $\theta \sim \mathcal{B}(p)$ with probability $p = d$ as our prior, which turns it into a bayesian neural network. Using MC Dropout [12], the stochastic nature of the resulting model can then be leveraged during inference to model the epistemic uncertainty of the network. For a pixel i of a testing input sample $\mathbf{x}$, this uncertainty can be quantified as the pixel-wise variance over n MC samples:

$$\text{Var}_i(\mathbf{x}) \approx \frac{1}{n} \sum_{k=1}^n f(\mathbf{x})_i^2 - \left(\frac{1}{n} \sum_{m=1}^n f(\mathbf{x})_i \right)^2 \quad (3)$$

, where $f(\mathbf{x})_i$ is the models response for pixel i .

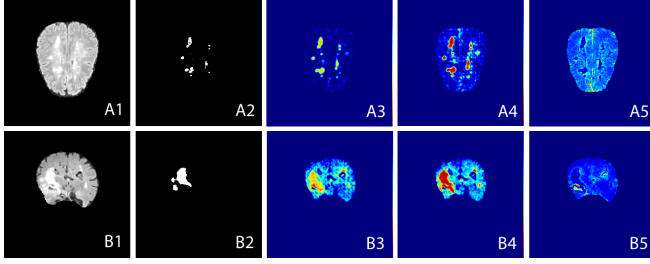


Fig. 2. Heatmap display for residuals and uncertainties. A1: input slice with MS lesions; A2: corresponding ground-truth segmentation; A3: the skipAE(#3) residual; A4: the skipBAE residual computed on the predictive mean; A5: the skipBAE pixel-wise uncertainty. B1-B5: The same for Glioblastoma.

2.4. Anomaly Detection

To leverage a trained non-bayesian model for UAD, we simply compute the residuals $r = |\mathbf{x} - \hat{\mathbf{x}}|$ for all axial slices of a testing subject. For our bayesian Skip-Autoencoder, we first aggregate a predictive mean of n MC reconstructions before the residuals are computed. Then, thresholding can be applied to produce a binary delineation. To reduce False Positives and to provide the physician with a smooth Precision-Recall-Curve from which an operating point t can be chosen, we further apply 3D bilateral filtering to the volume of residual images.

3. EXPERIMENTS AND RESULTS

3.1. Dataset

In our experiments we use a non-public dataset containing FLAIR and T1-weighted MR images of 50 healthy subjects, 49 subjects with MS lesions and 26 subjects with Glioblastoma. All data were acquired with a Philips Achieva 3T MR scanner and annotated by expert Neuro-radiologists. Informed consent was waived. In preparation, the T1 images were used for brain extraction using ROBEX [13] and for co-registration to bring all subjects and scans to the SRI24 ATLAS [14] space. For our experiments, we then only use co-registered FLAIR images, normalized into $[-1; 1]$. We utilize all non-empty axial FLAIR slices of the healthy subjects to train the models, where non-empty slices are determined from the brain extraction. From each pathology, we randomly pick three validation subjects each to determine an Operating Point (OP) for assessment and use all others for testing.

3.2. Metrics

For evaluation we compute the Precision-Recall-Curve (PRC) over all testing data and quantitatively compare methods by their area under this curve (AUPRC) as well as the patient-wise DICE-Score (determined at the most optimal

OP for the validation samples) across all testing patients. The AUPRC provides a meaningful measure for delineation (and thus detection) performance in the presence of heavy class-imbalance, which is apparent in both pathologies we experiment with.

3.3. Experiment A: Investigating Dropout

First, we trained three skipAEs with complete skip-connectivity at different rates of dropout $d \in \{0, 0.1, 0.5\}$ for 150 epochs to investigate the effect on identity-mitigation and UAD performance. We witness only subtle variation among their PRCs (see Fig. 3.3c), which implies that an identity mapping is not learned, regardless of dropout. Apparently, random weight initialization is enough to prevent identity.

3.4. Experiment B: Exploring Skip-Connections

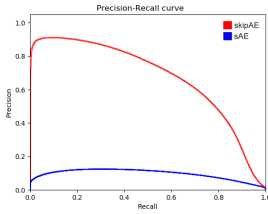
Afterwards, we trained a variety of skipAEs with different combinations of skip-connections at fixed d , again for 150 epochs. Since results from experiment A indicate that there is no necessity for dropout, we set $d = 0.0$. As a baseline, we first trained a spatial Autoencoder [2] (sAE) based on our skipAE architecture with all the skip-connections removed. To that, we compare models with either one, two or all skip-connections. Further, to determine if lower levels of the skipAE are not simply bypassed, we trained a model with skip-connections, but no bottleneck (z and $d3$ removed, see Table 1, #2), and a model which purely consists of only the first skip-connection (see Table 1, #6). We also compare to the skipBAE, i.e. the bayesian variant of our model following the work in [3]. The skipBAE aggregates a predictive mean from 100 MC samples at a dropout probability $p = 0.1$. All results are reported in Table 1. The best performance on MS was obtained with a single skip-connection close to the bottleneck (see Table. 1, #3). With increasingly excessive use of skip-connections, the performance on MS slightly drops, and slightly increases for Glioblastoma. In comparison to the sAE, the best skipAE yields an absolute improvement of 55% in AUPRC for detecting MS lesions and 17% for Glioblastoma. While still considerably better than the sAE, removal of the bottleneck (skipAE w/o z) leads to a reduction of performance in all measures for all pathologies. Interestingly, the skipBAE does not perform as good as the non-bayesian model.

3.5. Model Uncertainty in Bayesian Skip-Autoencoders

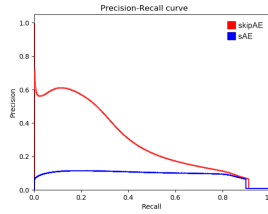
As a brief excursus, we leverage the stochastic nature of the skipBAE to compute the pixel-wise variance (see Eq. 3) on all testing slices and conduct a brief qualitative assessment of the uncertainty associated with anomalous pixels and normal tissue. Fig. 2 shows some heatmap displays of residuals produced by a skipAE (A3, B3), a skipBAE (A4, B4) as well as the pixel-wise variance for the latter (A5, B5). Blue depicts

Table 1. skipAE against alterations and baselines.

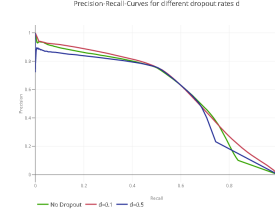
#	Model	Skip-Con.	MS lesions		Glioblastoma	
			AUPRC	DICE ($\mu \pm \sigma$)	AUPRC	DICE ($\mu \pm \sigma$)
1	sAE[2]	-	0.0956	0.1936 ± 0.1177	0.0933	0.2043 ± 0.1928
2	skipAE w/o z	s3	0.3139	0.4000 ± 0.1738	0.2760	0.3020 ± 0.2065
3	skipAE	s3	0.6398	0.5822 ± 0.1937	0.2924	0.3057 ± 0.1765
4	skipAE	s2,s3	0.6102	0.5714 ± 0.1852	0.2988	0.3067 ± 0.1733
5	skipAE	s0,s1,s2,s3	0.5909	0.5520 ± 0.1857	0.3056	0.3147 ± 0.1663
6	skipAE	s0	0.5906	0.5461 ± 0.1893	0.2899	0.3269 ± 0.1905
7	skipBAE[3]	s3	0.6275	0.5539 ± 0.2103	0.2767	0.3171 ± 0.1720



a) PRCs on MS lesion testing set



b) PRCs on Glioblastoma testing set



c) PRCs on MS at different dropout rates

Fig. 3. PRCs of the sAE vs. skipAE(#3) on two different pathologies and among skipAEs(#5) with different dropout rates.

low and yellow/orange/red reflects high uncertainty. Close inspection reveals that pixels of hyperintense anomalies (such as MS lesions) have low variance compared to normal tissue. It is highest around sharp boundaries and complex anatomical structures such as gyri and sulci.

3.6. Discussion

Our ablative study shows that skip-connections in AEs are very helpful for UAD, in particular for the detection of hyperintense anomalies such as MS lesions. Our skipAE outperforms the spatial AE by a large margin on both pathologies. At the same time, we also show that sacrificing the bottleneck or any lower layers of the skipAE impairs UAD performance, which implies that it benefits heavily from the multi-scale feature extraction. The number of skip-connections needs to be carefully selected, as their excessive use enables the reconstruction of anomalies as well, though. On hyperintense lesions, we find the skipBAE to be inferior to the skipAE, because averaging over the MC samples induces slight blurriness in the reconstructions. Unfortunately, the AE-based detection of Glioblastoma does not work as good as the detection of MS lesions. Glioblastoma are morphologically and intensity-wise much more diverse than purely hyperintense lesions and thus benefit from more skip-connections, but the non-hyperintense tumor areas are often reconstructed too well to be recognized from the residuals.

4. CONCLUSION

In conclusion, we showed that skip-connections in AEs lead to considerable improvements in hyperintense anomaly detection in brain MR data. The skip-connections enable high fidelity reconstructions, which are key for UAD in brain MRI. At no time, an identity mapping was learned, even if dropout was disabled. However, if dropout is active, the resulting model allows to estimate epistemic uncertainty. Our brief excursus to epistemic uncertainty revealed that residuals vary mainly for healthy tissue, and hyperintense anomalous areas tend to have low uncertainty. We leave more thorough investigations of the role of uncertainty in UAD to future work, though.

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